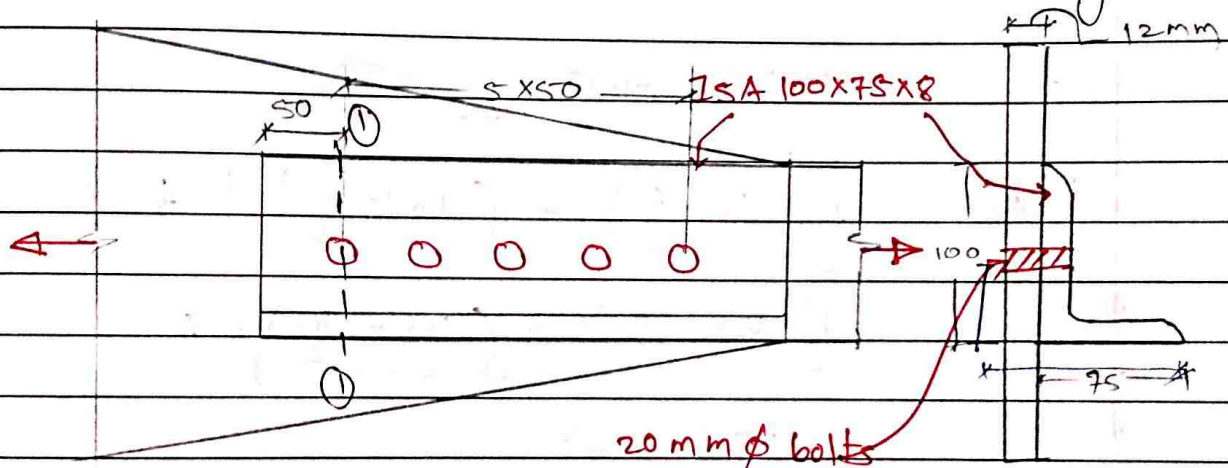


Q.1] A single unequal angle $100 \times 75 \times 8$ mm. is connected to a 12 mm thick G.P. at the ends with 6 nos. 20 mm dia. bolts to transfer tension. Determine the design tensile strength of the angle if the yield & the ultimate stress of the steel used are 250 MPa & 420 MPa.

a) If the gusset is connected to 100 mm leg.

b) If the G.P. is connected to 75 mm leg.



→ Given :-

Step 1] Given data.

$$f_y = 250 \text{ MPa} \text{ \& } f_u = 420 \text{ MPa}$$

$$d = 20 \text{ mm.}$$

$$\therefore d_n = 20 + 2 = 22 \text{ mm.}$$

$$p = 50 \text{ mm, } e_d = 50 \text{ mm.}$$

$$Z_{mo} = 1.1 \text{ \& } Z_{m1} = 1.25$$

Case I] The 100 mm leg bolted to the G.P. :-

$$A_{nc} = \left(b_1 - \frac{t}{2} \right) t \text{ - deduction}$$

$$\begin{aligned}
 A_{nc} &= (l_1 - t/2) t - n d t \\
 &= (100 - 8/2) \times 8 - 22 \times 8 \\
 &= 592 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 A_{g(out)} &= (l_2 - t/2) \times t \\
 &= (75 - 8/2) \times 8 \\
 &= 568 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 A_g &= \left[\left(l_1 - \frac{t}{2} \right) + \left(l_2 - \frac{t}{2} \right) \right] t \\
 &= \left[\left(100 - \frac{8}{2} \right) + \left(75 - \frac{8}{2} \right) \right] \times 8 \\
 &= 1336 \text{ mm}^2
 \end{aligned}$$

(i) Strength is governed by yielding of gross section.

$$T_{dy} = \frac{A_g f_y}{\gamma_{mo}} = \frac{1336 \times 250}{1.1}$$

$$T_{dy} = 303636 \text{ N} = \underline{\underline{303.63 \text{ kN}}}$$

(ii) Strength is governed by rupture of critical section.

$$T_{dn} = 0.9 f_u A_{nc} / \gamma_{m1} + \beta \times A_{go} \times \frac{f_y}{\gamma_{m0}}$$

$$\begin{aligned}
 \beta &= 1.4 - 0.076 \left(\frac{w}{t} \right) \left(\frac{f_y}{f_u} \right) \left(\frac{b_s}{L_c} \right) \\
 &= 1.4 - 0.076 \left(\frac{75}{8} \right) \left(\frac{250}{420} \right) \left(\frac{117}{250} \right) \\
 &= 0.839 \quad 1.2 \\
 &= 0.84 \quad 1.2 \geq 0.7 \\
 &\quad < 2.47
 \end{aligned}$$

$$w = 75 \text{ mm}, \quad t = 8 \text{ mm}, \quad L_c = 5 \times 50 = 250.$$

$$\begin{aligned}
 b_s &= w + w_1 - t \\
 &= 75 + 50 - 8
 \end{aligned}$$

$$b_s = 117 \text{ mm}$$

$$\therefore T_{dn} = \frac{0.9 \times 420 \times 592}{1.25} + \frac{0.84 \times 568 \times 250}{1.01}$$

$$\therefore T_{dn} = 333.93 \text{ kN.}$$

$$\begin{aligned}
 \text{i)} \quad L_{vg} &= 5 \times 50 + 50 \\
 &= 300 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad L_{vn} &= L_{vg} - 5.5 \times d_h \\
 &= 300 - 5.5 \times 22 \\
 &= 179 \text{ mm}
 \end{aligned}$$

$$\text{iii)} \quad L_{tg} = e_d = 50 \text{ mm.}$$

$$\begin{aligned}
 \text{iv)} \quad L_{tn} &= L_{tg} - \frac{1}{2} \times d_h \\
 &= 50 - 0.5 \times 22 \\
 &= 39 \text{ mm}
 \end{aligned}$$

iii) Strength governed by Block shear (T_{db}):

$$i) A_{vg} = L_{vg} \times t = 300 \times 8 = 2400 \text{ mm}^2$$

$$ii) A_{vn} = 179 \times 8 = 1432 \text{ mm}^2$$

$$iii) A_{tg} = L_{tg} \times t = 50 \times 8 = 400 \text{ mm}^2$$

$$iv) A_{tn} = L_{tn} \times t = 39 \times 8 = 312 \text{ mm}^2$$

$$T_{db} = \frac{0.577 \times f_y A_{vg}}{\gamma_{mo}} + \frac{0.9 \times f_u \times A_{tn} \times f_y}{f_y \gamma_{m1}}$$

$$= \frac{0.577 \times 250 \times 2400}{1.1} + \frac{0.9 \times 420 \times 312 \times 250}{1.25}$$

$$= 409.076 \text{ kN.}$$

$$T_{db} = \frac{0.52 \times f_u \times A_{vn}}{\gamma_{m1}} + \frac{A_{tg} f_y}{\gamma_{mo}}$$

$$= \frac{0.52 \times 420 \times 1432}{1.25} + \frac{400 \times 250}{1.1}$$

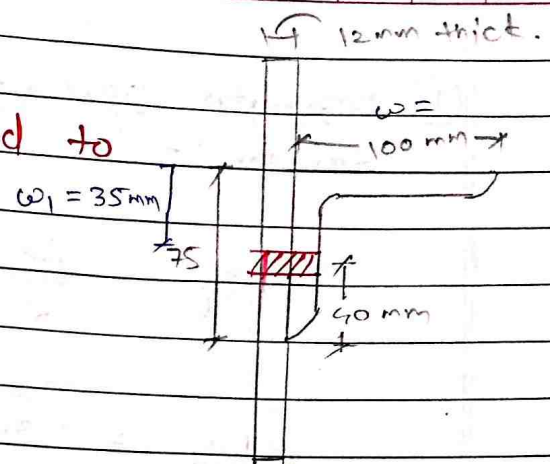
$$T_{db} = 341.108 \text{ kN.}$$

T_{db} is min. of Both.

$$T_{db} = 341.108 \text{ kN.}$$

Case 2]:

75 mm leg is connected to G.P.



$$\begin{aligned}
 A_{nc} &= \left(d_1 - \frac{t}{2} \right) \times t - n d_n \\
 &= (75 - 4) \times 8 - 22 \times 8 \\
 &= 392 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 A_{g(out)} &= \left(d_2 - \frac{t}{2} \right) \times t \\
 &= (100 - 4) \times 8 \\
 &= 768 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 A_g &= \left[\left(d_1 - \frac{t}{2} \right) + \left(d_2 - \frac{t}{2} \right) \right] \times t \\
 &= \left[(75 - 4) + (100 - 4) \right] \times 8
 \end{aligned}$$

$$= 1336 \text{ mm}^2$$

① Strength governed by yielding of gross section:-

$$T_{dy} = \frac{A_g f_y}{\gamma_{mo}} = \frac{1336 \times 250}{1.1}$$

$$T_{dy} = 303.63 \text{ kN}$$

②

② Strength as governed by rupture of critical section.

$$w = 100 \text{ mm}, w_1 = 35 \text{ mm}, t = 8 \text{ mm}$$

$$L_c = 5 \times 50 = 250 \text{ mm}$$

$$\begin{aligned} b_s &= w + w_1 - t \\ &= 100 + 35 - 8 \\ &= 127 \text{ mm} \end{aligned}$$

$$\beta = 1.4 - 0.076 \left(\frac{w}{t} \right) \left(\frac{f_y}{f_u} \right) \left(\frac{b_s}{L_c} \right)$$

$$= 1.4 - 0.076 \left(\frac{100}{8} \right) \left(\frac{250}{420} \right) \left(\frac{127}{250} \right)$$

$$= 1.112 > 0.7.$$

$$= 1.112 < \left(\frac{f_u \gamma_{mo}}{f_y \gamma_{m1}} \right) = \frac{2.31}{1.47}.$$

$$\therefore T_{dn} = \frac{0.9 f_u A_{nc}}{2m_1} + \frac{\beta \times A_{gv} \times f_y}{2m_2}$$

$$= \frac{0.9 \times 420 \times 392}{1.25} + \frac{1.112 \times 768 \times 250}{1.1}$$

$$\therefore T_{dn} = 312.63 \text{ kN.}$$

③ Strength governed by Block shear (T_{db})

$$i) L_{vg} = 5 \times 50 + 80 = 330 \text{ mm.}$$

$$ii) L_{vn} = L_{vg} - 5.5 \times d_h$$

$$= 330 - 5.5 \times 22$$

$$= 179 \text{ mm}$$

$$iii) L_{tg} = e_d = 40 \text{ mm}$$

$$iv) L_{tn} = L_{tg} - \frac{1}{2} d_h$$

$$= 40 - 0.5 \times 22$$

$$= 29 \text{ mm}$$

$$① A_{vg} = 2400 \text{ mm}^2.$$

$$② A_{vn} = 1432 \text{ mm}^2$$

$$③ A_{tg} = 400 \text{ mm}^2 \quad 40 \times 8 = 320 \text{ mm}^2$$

$$④ A_{tn} = 312 \text{ mm}^2 \quad 29 \times 8 = 232 \text{ mm}^2$$

$$T_{db} = \frac{0.577 \times f_y \times A_{vg}}{\gamma_{mo}} + \frac{0.9 \times f_u \times A_{tn}}{\gamma_{m1}}$$

$$= \frac{0.577 \times 250 \times 2400}{1.1} + \frac{0.9 \times 420 \times 2312}{1.25}$$

$$= 409.07 \text{ KN} \quad 384.88 \text{ KN}$$

$$T_{db} = \frac{0.52 \times f_u \times A_{vn}}{\gamma_{m1}} + \frac{A_{tg} \times f_y}{\gamma_{mo}}$$

$$= \frac{0.52 \times 420 \times 1432}{1.25} + \frac{320 \times 250}{1.1}$$

$$T_{db} = 341.108 \text{ KN}$$

$$= 322.92 \text{ KN}$$

T_{db} is min of Both.

$$\therefore T_{db} = \begin{matrix} 341.108 \text{ KN} \\ 322.92 \end{matrix}$$

$\therefore T_d$ is min. of all 3 cases values.

$$T_d = 303.63 \text{ KN}$$